

A Formula Atlas for the Thue–Morse Sequence

Exact arithmetic, Pascal-triangle, product, Prouhet,
Fourier, spectral, infinite-product, and Fabius representations

A consolidated and expanded version of four research drafts

in Vladimir Reshetnikov’s PROVEIT project

This manuscript merges the four files *Further Formulae for the Thue–Morse Sequence–1* through *–4*. Repeated statements have been collapsed, notation has been standardized, proofs have been cross-checked, and the material has been arranged as a dependency-respecting atlas rather than as four parallel reports. New material includes a minimal-kernel viewpoint, a Catalan-valuation formula, a direct natural-boundary proof, entire shifted Dirichlet series, additional Boolean-lattice inversions, word-complexity consequences, a base- B root-of-unity Prouhet theorem, and a modern distribution survey. No claim of historical priority is intended.

26 August 2026

Abstract

Let $\tau(n) = \text{wt}(n) \bmod 2$ be the zero–one Thue–Morse bit and $\varepsilon_n = (-1)^{\text{wt}(n)} = 1 - 2\tau(n)$ its signed form. Almost every useful formula becomes transparent after moving among five equivalent objects: binary digit parity, a character of the Boolean cube, the finite product $\prod_{j < m} (1 - z^{2^j})$, the substitution $0 \mapsto 01, 1 \mapsto 10$, and a mixed finite-difference operator. The purpose of this document is to make those translations explicit and reusable. It develops arithmetic and Pascal identities, exact prefix discrepancy, valuation recurrences, partition and determinant formulas, Prouhet cancellation and all-order moments, ordinary and Walsh Fourier transforms, Riesz products and autocorrelation, Mahler functions, Dirichlet and product transforms, finite-state and word combinatorics, base- B extensions, and the discrete bridge to the Fabius function. Results imported from the literature are labelled as such; all other displayed identities are proved or reduced to a previously proved identity in the text.

Contents

1	Conventions, provenance, and a one-page map	1
1.1	Formula map	1
2	Binary recursion, blocks, automata, and uniqueness	1
2.1	The two-scale recurrence and substitution	1
2.2	Binary concatenation	2
2.3	The exact two-kernel and a minimal automaton	2
2.4	Nonperiodicity	3
3	Floor, fractional-part, trigonometric, and bitwise formulas	3
3.1	Digit extraction	3
3.2	A sign product and an entire sinc product	3
3.3	XOR and carry-free multiplication	4
4	Two-adic valuations, carries, and Catalan numbers	4
4.1	Legendre and the central binomial coefficient	4
4.2	Catalan valuation	5
4.3	Kummer carries and multinomials	5
5	Pascal-triangle formulas and Boolean-lattice inversion	5
5.1	Odd entries and the binomial–logarithm formula	5
5.2	A log-free residue formula	6
5.3	Row enumerators, powers-of-two columns, and submasks	6
5.4	Pascal parity as a zeta matrix	7
5.5	Odd multinomial support	7

6	Exact balance, interval discrepancy, and dyadic blocks	8
7	Finite and infinite Euler products	8
7.1	The finite mother product	8
7.2	The infinite product and Mahler equation	9
7.3	A direct natural-boundary theorem	9
8	The ruler sequence, coefficient recurrences, partitions, and determinants	10
8.1	Partition and Bell-polynomial formulas	10
8.2	A factorial determinant	11
9	Prouhet cancellation and a master finite-difference identity	11
9.1	Exact vanishing order	11
9.2	The operator formula	12
9.3	All higher moments	12
10	Sparse Prouhet identities on arbitrary Pascal supports	13
11	The bit series in characteristic two	14
12	Iterated prefix sums and terminal zero runs	14
13	Walsh, ordinary Fourier, and cyclotomic formulas	15
13.1	The Boolean-cube character	15
13.2	Trigonometric factorization	15
13.3	Exact discrete Fourier transform	16
14	Riesz products, autocorrelation, and spectral type	16
15	Base-b Thue–Morse constants and Mahler transcendence	17
16	Shifted Dirichlet series and Mellin transforms	17
17	A two-parameter product calculus	18
17.1	The Woods–Robbins product	19
17.2	Three dyadic block-product limits	19
18	Combinatorics on the infinite word	19
18.1	Uniform recurrence and reversal symmetry	19
18.2	An elementary linear complexity bound	20

19 Distribution in arithmetic and polynomial subsequences	20
20 Base-B root-of-unity generalization	21
21 The discrete bridge to the Fabius and up-functions	22
22 Representation comparison and a Lean roadmap	23
A The first 64 terms	23
B A compact identity index	24
C Executable checks	24
D Bibliographic notes	25

1 Conventions, provenance, and a one-page map

Throughout $\mathbb{N} = \{0, 1, 2, \dots\}$. If

$$n = \sum_{j \geq 0} b_j(n) 2^j, \quad b_j(n) \in \{0, 1\},$$

then

$$\text{wt}(n) = \sum_{j \geq 0} b_j(n), \quad \tau(n) = \text{wt}(n) \bmod 2, \quad \varepsilon_n = (-1)^{\text{wt}(n)} = 1 - 2\tau(n). \quad (1.1)$$

The terminology *evil* and *odious* refers respectively to $\tau(n) = 0$ and $\tau(n) = 1$.

The four source drafts were independently organized and consequently overlap in their basic recurrences, Lucas/Kummer formulas, finite products, Prouhet identities, Fourier products, prefix sums, and Woods–Robbins evaluation. The present text retains one proof of each central theorem and places specialized consequences immediately after the theorem that generates them. The source files are available at

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts>.

1.1 Formula map

Viewpoint	Exact representation	Main use
Binary recursion	$\varepsilon_{2n} = \varepsilon_n, \varepsilon_{2n+1} = -\varepsilon_n$	Fast computation, blocks, automata
Boolean character	$\varepsilon_n = (-1)^{\mathbf{1} \cdot \mathbf{b}(n)}$	XOR, Walsh transform, submasks
Valuation	$\text{wt}(n) = \nu_2 \binom{2n}{n}$	Binomial/Catalan formulas
Pascal row	$\#\{k : \binom{n}{k} \text{ odd}\} = 2^{\text{wt}(n)}$	Binomial-log and log-free formulas
Finite product	$P_m(z) = \prod_{j < m} (1 - z^{2^j})$	Moments, DFT, Riesz products
Infinite product	$E(z) = \prod_{j \geq 0} (1 - z^{2^j})$	Mahler equation, constants, transforms
Finite difference	$\sum_{n < 2^m} \varepsilon_n f(n) = (-1)^m \Delta_1 \cdots \Delta_{2^{m-1}} f(0)$	Prouhet, remainders, reciprocal sums
Prefix transform	$\sum_n S_n^{(k)} z^n = E(z)/(1 - z)^k$	Discrete antiderivatives, Fabius bridge
Spectral measure	$2^{-m} P_m(e^{2\pi i x}) ^2 dx$	Autocorrelation, singular spectrum

Remark 1.1 (Proof status). A label “Literature theorem” means that only context or a proof sketch is given. “Added in this synthesis” means absent from the four source drafts as a named result; it does not assert novelty in the historical literature. The Lean file names mentioned near the end are a formalization roadmap, not a claim that every identity in this atlas is already formalized.

2 Binary recursion, blocks, automata, and uniqueness

2.1 The two-scale recurrence and substitution

Theorem 2.1 (Fundamental recursion). *For every $n \geq 0$,*

$$\boxed{\varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n,} \quad (2.1)$$

and equivalently

$$\tau(2n) = \tau(n), \quad \tau(2n+1) = 1 - \tau(n). \quad (2.2)$$

Proof. Appending a binary zero does not change the digit sum, while appending a binary one increases it by one. Taking parity gives (2.1) and (2.2). $\square$

Starting from 0, the morphism

$$\mu(0) = 01, \quad \mu(1) = 10 \quad (2.3)$$

produces the zero–one word. In signed notation the corresponding operation replaces a block W by $W, -W$. Thus the block of length 2^{m+1} is the block of length 2^m followed by its complement.

2.2 Binary concatenation

Theorem 2.2 (Block multiplicativity). *If $m, q \geq 0$ and $0 \leq r < 2^m$, then*

$$\boxed{\varepsilon_{2^m q + r} = \varepsilon_q \varepsilon_r}, \quad \tau(2^m q + r) \equiv \tau(q) + \tau(r) \pmod{2}. \quad (2.4)$$

Proof. The binary digits of $2^m q + r$ are the digits of q , shifted by m places, followed by the m -digit expansion of r ; the supports are disjoint, so their weights add. $\square$

In particular,

$$(\varepsilon_{2^m}, \dots, \varepsilon_{2^{m+1}-1}) = -(\varepsilon_0, \dots, \varepsilon_{2^m-1}). \quad (2.5)$$

2.3 The exact two-kernel and a minimal automaton

Theorem 2.3 (Exact two-kernel). *For $e \geq 0$ and $0 \leq r < 2^e$,*

$$\varepsilon_{2^e n + r} = \varepsilon_r \varepsilon_n. \quad (2.6)$$

Consequently the 2-kernel of the signed sequence consists of exactly the two sequences $(\varepsilon_n)_{n \geq 0}$ and $(-\varepsilon_n)_{n \geq 0}$.

Proof. Apply equation (2.4). The sign is $\varepsilon_r \in \{\pm 1\}$. Both possibilities occur, for example at $r = 0$ and $r = 1$. $\square$

Added in this synthesis. The preceding theorem makes the minimal automaton completely explicit. Let

$$M_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad M_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad e_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

If $n = \sum_{j=0}^{m-1} b_j 2^j$, then

$$v(n) = M_{b_0} \cdots M_{b_{m-1}} e_0 = \begin{pmatrix} 1 - \tau(n) \\ \tau(n) \end{pmatrix}, \quad (2.7)$$

so

$$\tau(n) = \begin{pmatrix} 0 & 1 \end{pmatrix} v(n), \quad \varepsilon_n = \begin{pmatrix} 1 & -1 \end{pmatrix} v(n). \quad (2.8)$$

The two states record only the parity already read; input 1 toggles the state and input 0 fixes it. Two states are necessary because the outputs at 0 and 1 differ.

2.4 Nonperiodicity

Theorem 2.4 (No eventual period). *The Thue–Morse sequence is not eventually periodic.*

Proof. Suppose that $p > 0$ were an eventual period. Fix $q \geq 0$ and choose k so large that $2^k + q$ and $2^k + q + p$ lie in the periodic tail and $q + p < 2^k$. Block multiplicativity gives

$$-\varepsilon_q = \varepsilon_{2^k+q} = \varepsilon_{2^k+q+p} = -\varepsilon_{q+p},$$

so p would in fact be a period from the beginning. Choose k with $2^k > p$. Periodicity would imply $\varepsilon_{2^k-p} = \varepsilon_{2^k} = -1$. Yet $2^k - p = (2^k - 1) - (p - 1)$, so its k digits are the complements of the k digits of $p - 1$, and

$$\varepsilon_{2^k-p} = (-1)^{k-\text{wt}(p-1)}.$$

This alternates when k is increased by one, a contradiction. $\square$

3 Floor, fractional-part, trigonometric, and bitwise formulas

3.1 Digit extraction

For every $j \geq 0$,

$$b_j(n) = \left\lfloor \frac{n}{2^j} \right\rfloor - 2 \left\lfloor \frac{n}{2^{j+1}} \right\rfloor. \quad (3.1)$$

Therefore, with all but finitely many factors equal to 1,

$$\varepsilon_n = \prod_{j \geq 0} \left(1 - 2 \left\lfloor \frac{n}{2^j} \right\rfloor + 4 \left\lfloor \frac{n}{2^{j+1}} \right\rfloor \right). \quad (3.2)$$

Also,

$$\text{wt}(n) = n - \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor = \sum_{j \geq 1} \left\{ \frac{n}{2^j} \right\}, \quad (3.3)$$

where $\{x\} = x - \lfloor x \rfloor$. The second equality follows because $\sum_{j \geq 1} n/2^j = n$.

Consequently,

$$\varepsilon_n = (-1)^{n + \sum_{j \geq 1} \lfloor n/2^j \rfloor} = \cos \left(\pi \sum_{j \geq 1} \left\{ \frac{n}{2^j} \right\} \right), \quad \tau(n) = \frac{1 - \varepsilon_n}{2}. \quad (3.4)$$

3.2 A sign product and an entire sinc product

Proposition 3.1 (Rademacher–sine sign product). *For every $n \geq 0$,*

$$\varepsilon_n = \prod_{j \geq 0} \text{sgn} \sin \left(\frac{\pi(n + \frac{1}{2})}{2^j} \right). \quad (3.5)$$

Only finitely many factors are negative.

Proof. The sign of the j -th sine is $(-1)^{\lfloor (n+1/2)/2^j \rfloor} = (-1)^{\lfloor n/2^j \rfloor}$. Multiplying and using (3.3) modulo two gives $(-1)^{\text{wt}(n)}$. $\square$

Added in this synthesis. Define $\text{sinc } y = \sin y/y$, with $\text{sinc } 0 = 1$, and

$$\Phi(x) = \prod_{j \geq 0} \text{sinc}\left(\frac{\pi x}{2^j}\right). \quad (3.6)$$

The product converges locally uniformly and hence defines an entire function. For every integer $N \geq 0$ and $N < x < N + 1$,

$$\boxed{\text{sgn } \Phi(x) = \varepsilon_N.} \quad (3.7)$$

Indeed, the denominator of each factor is positive and the signs of the numerators multiply exactly as in (3.5), with x in place of $N + 1/2$. Thus the Thue–Morse word is the sign itinerary of a single canonical entire product between its successive integer zeros.

3.3 XOR and carry-free multiplication

For nonnegative integers a, b ,

$$\text{wt}(a \text{ xor } b) = \text{wt}(a) + \text{wt}(b) - 2 \text{wt}(a \text{ and } b). \quad (3.8)$$

Hence

$$\boxed{\varepsilon_{a \text{ xor } b} = \varepsilon_a \varepsilon_b, \quad \tau(a \text{ xor } b) \equiv \tau(a) + \tau(b) \pmod{2}.} \quad (3.9)$$

This says that $n \mapsto \varepsilon_n$, restricted to m -bit integers, is the Walsh character indexed by the all-ones vector.

If a and $b = 0$, then $a + b = a \text{ xor } b$, and therefore

$$\varepsilon_{a+b} = \varepsilon_a \varepsilon_b. \quad (3.10)$$

The correction in the presence of carries will appear in [section 4](#).

4 Two-adic valuations, carries, and Catalan numbers

4.1 Legendre and the central binomial coefficient

Legendre’s formula in base two is

$$\nu_2(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{2^j} \right\rfloor = n - \text{wt}(n). \quad (4.1)$$

It immediately yields one of the cleanest arithmetic encodings.

Theorem 4.1 (Central-binomial valuation). *For every $n \geq 0$,*

$$\boxed{\nu_2\binom{2n}{n} = \text{wt}(n), \quad \varepsilon_n = (-1)^{\nu_2\binom{2n}{n}}.} \quad (4.2)$$

Proof. By (4.1),

$$\nu_2\binom{2n}{n} = (2n - \text{wt}(2n)) - 2(n - \text{wt}(n)) = \text{wt}(n),$$

because $\text{wt}(2n) = \text{wt}(n)$. □

4.2 Catalan valuation

Theorem 4.2 (Catalan-number formula). *Added in this synthesis.* Let $C_n = \frac{1}{n+1} \binom{2n}{n}$. Then

$$\boxed{\nu_2(C_n) = \text{wt}(n+1) - 1 = \text{wt}(n) - \nu_2(n+1).} \quad (4.3)$$

Consequently

$$\varepsilon_{n+1} = -(-1)^{\nu_2(C_n)}, \quad \varepsilon_n = (-1)^{\nu_2(C_n) + \nu_2(n+1)}. \quad (4.4)$$

Proof. Use (4.2) and subtract $\nu_2(n+1)$. When one is added to n , its trailing block of $\nu_2(n+1)$ one-bits becomes zero and the next zero becomes one, so $\text{wt}(n+1) = \text{wt}(n) + 1 - \nu_2(n+1)$. $\square$

4.3 Kummer carries and multinomials

Kummer's theorem gives

$$\nu_2 \binom{a+b}{a} = \text{wt}(a) + \text{wt}(b) - \text{wt}(a+b), \quad (4.5)$$

which is the number of binary carries, counted with multiplicity through propagation. Taking parity yields the exact carry correction

$$\boxed{\varepsilon_{a+b} = \varepsilon_a \varepsilon_b (-1)^{\nu_2 \binom{a+b}{a}}.} \quad (4.6)$$

More generally, if $k_1 + \dots + k_r = n$, then

$$\nu_2 \binom{n}{k_1, \dots, k_r} = \sum_{i=1}^r \text{wt}(k_i) - \text{wt}(n), \quad (4.7)$$

and therefore

$$\boxed{\varepsilon_n = \left(\prod_{i=1}^r \varepsilon_{k_i} \right) (-1)^{\nu_2 \binom{n}{k_1, \dots, k_r}}.} \quad (4.8)$$

5 Pascal-triangle formulas and Boolean-lattice inversion

5.1 Odd entries and the binomial–logarithm formula

Lucas's theorem modulo two says

$$\binom{n}{k} \equiv 1 \pmod{2} \iff k \preceq_2 n, \quad (5.1)$$

meaning that each one-bit of k is also a one-bit of n . Hence the number of odd entries in row n of Pascal's triangle is

$$\boxed{\# \left\{ 0 \leq k \leq n : \binom{n}{k} \text{ is odd} \right\} = 2^{\text{wt}(n)}.} \quad (5.2)$$

Set

$$X(n) = \sum_{k=0}^n (-1)^{\binom{n}{k}}. \quad (5.3)$$

Every even entry contributes $+1$, every odd entry -1 , so

$$\boxed{X(n) = n + 1 - 2^{\text{wt}(n)+1}.} \quad (5.4)$$

This recovers the source document's binomial–logarithm formula:

$$\tau(n) = \frac{1 + (-1)^{\log_2(n+1-X(n))}}{2}, \quad (5.5)$$

because $n + 1 - X(n) = 2^{\text{wt}(n)+1}$. The logarithm is exact, not numerical.

5.2 A log-free residue formula

Let

$$S(n) = 1 + X(n) = n + 2 - 2^{\text{wt}(n)+1}. \quad (5.6)$$

Since powers of two alternate between 1 and 2 modulo three,

$$2n + S(n) \equiv \begin{cases} 0 & (\text{mod } 3), \quad \text{wt}(n) \text{ even,} \\ 1 & (\text{mod } 3), \quad \text{wt}(n) \text{ odd.} \end{cases} \quad (5.7)$$

Thus the least nonnegative residue already is a bit:

$$\boxed{\tau(n) = [2n + 1 + \sum_{k=0}^n (-1)^{\binom{n}{k}}]_3.} \quad (5.8)$$

Equivalently, because the residue can only be 0 or 1,

$$\tau(n) = \frac{2}{\sqrt{3}} \sin \left(\frac{2\pi}{3} \left(2n + 1 + \sum_{k=0}^n (-1)^{\binom{n}{k}} \right) \right). \quad (5.9)$$

5.3 Row enumerators, powers-of-two columns, and submasks

For an indeterminate u , define the parity enumerator

$$R_n(u) = \sum_{k=0}^n u^{\binom{n}{k} \bmod 2}.$$

Then

$$\boxed{R_n(u) = n + 1 + (u - 1)2^{\text{wt}(n)}.} \quad (5.10)$$

Specializing $u = -1$ gives (5.4).

Lucas's theorem at the power-of-two columns reads

$$\binom{n}{2^j} \bmod 2 = b_j(n). \quad (5.11)$$

Therefore

$$\boxed{\varepsilon_n = \prod_{j \geq 0} (-1)^{\binom{n}{2^j}}.} \quad (5.12)$$

The complete submask enumerator is

$$\boxed{\sum_{k \preceq_2 n} z^{\text{wt}(k)} = (1 + z)^{\text{wt}(n)}}. \quad (5.13)$$

At $z = -2$,

$$\varepsilon_n = \sum_{k \preceq_2 n} (-2)^{\text{wt}(k)}. \quad (5.14)$$

Combining this with (4.2) and the indicator for oddness produces a formula involving only binomial coefficients and two-adic valuations:

$$\boxed{\varepsilon_n = \sum_{k=0}^n \frac{1 - (-1)^{\binom{n}{k}}}{2} (-2)^{\nu_2 \binom{2k}{k}}}. \quad (5.15)$$

5.4 Pascal parity as a zeta matrix

Let

$$P_{n,k} = \mathbf{1}_{k \preceq_2 n} = \binom{n}{k} \bmod 2. \quad (5.16)$$

On every finite initial Boolean lattice, P is the zeta matrix. Its inverse is the Möbius matrix

$$(P^{-1})_{n,k} = P_{n,k} (-1)^{\text{wt}(n) - \text{wt}(k)}. \quad (5.17)$$

The first column is precisely

$$(P^{-1})_{n,0} = (-1)^{\text{wt}(n)} = \varepsilon_n. \quad (5.18)$$

This places the signed Thue–Morse sequence inside the elementary incidence algebra of the Boolean lattice.

5.5 Odd multinomial support

For fixed $r \geq 2$, an r -nomial coefficient $\binom{n}{k_1, \dots, k_r}$ is odd exactly when each one-bit of n is assigned to exactly one of the r parts. Hence

$$\boxed{\# \left\{ (k_1, \dots, k_r) : \sum k_i = n, \binom{n}{k_1, \dots, k_r} \text{ odd} \right\} = r^{\text{wt}(n)}}. \quad (5.19)$$

Since the total number of weak compositions is $\binom{n+r-1}{r-1}$,

$$\sum_{k_1 + \dots + k_r = n} (-1)^{\binom{n}{k_1, \dots, k_r}} = \binom{n+r-1}{r-1} - 2r^{\text{wt}(n)}. \quad (5.20)$$

For any odd $r \equiv -1 \pmod{4}$, the parity of $\text{wt}(n)$ can again be recovered from a small residue of the right-hand side; the case $r = 3$ is the multinomial analogue of (5.8).

6 Exact balance, interval discrepancy, and dyadic blocks

Define the half-open signed prefix discrepancy

$$\mathcal{D}(N) = \sum_{0 \leq n < N} \varepsilon_n. \quad (6.1)$$

Theorem 6.1 (Exact prefix discrepancy). *For every $q \geq 0$,*

$$\boxed{\mathcal{D}(2q) = 0, \quad \mathcal{D}(2q + 1) = \varepsilon_q.} \quad (6.2)$$

In particular $|\mathcal{D}(N)| \leq 1$.

Proof. The first $2q$ terms cancel in pairs because $\varepsilon_{2j} + \varepsilon_{2j+1} = 0$. The next term is $\varepsilon_{2q} = \varepsilon_q$. $\square$

Let $A(N) = \sum_{n < N} \tau(n)$ count odious indices in the prefix. Since $\tau = (1 - \varepsilon)/2$,

$$\boxed{A(2q) = q, \quad A(2q + 1) = q + \tau(q).} \quad (6.3)$$

Thus every even prefix is exactly balanced.

For $0 \leq a \leq b$,

$$\sum_{n=a}^{b-1} \varepsilon_n = \mathcal{D}(b) - \mathcal{D}(a), \quad \left| \sum_{n=a}^{b-1} \varepsilon_n \right| \leq 2. \quad (6.4)$$

Consequently the number of ones in any interval differs from half the interval length by at most one. The signed bound two is sharp, for example on $[1, 3)$.

Block multiplicativity strengthens this to an exact local formula. If $0 \leq r \leq 2^m$, then

$$\boxed{\sum_{j=0}^{r-1} \varepsilon_{2^m q + j} = \varepsilon_q \mathcal{D}(r).} \quad (6.5)$$

Every aligned block of length 2^m , $m \geq 1$, has zero signed sum, and every partial block has discrepancy zero or one up to a global sign.

The ordinary generating series of the discrepancy is especially sparse:

$$\sum_{N \geq 0} \mathcal{D}(N) z^N = \frac{z}{1-z} E(z) = z E(z^2) = z \prod_{j \geq 1} (1 - z^{2^j}), \quad |z| < 1, \quad (6.6)$$

where E is introduced in [section 7](#). Coefficient comparison recovers (6.2).

7 Finite and infinite Euler products

7.1 The finite mother product

For $m \geq 0$, define

$$P_m(z) = \sum_{n=0}^{2^m-1} \varepsilon_n z^n. \quad (7.1)$$

Theorem 7.1 (Finite Euler product).

$$P_m(z) = \prod_{j=0}^{m-1} (1 - z^{2^j}). \quad (7.2)$$

Proof. When the product is expanded, choosing $-z^{2^j}$ from the factors indexed by a subset J produces the unique exponent $n = \sum_{j \in J} 2^j$ with coefficient $(-1)^{|J|} = \varepsilon_n$. $\square$

The recursion

$$P_{m+1}(z) = (1 - z^{2^m})P_m(z) \quad (7.3)$$

is the generating-polynomial form of block complementation. If $d = 2^m - 1$, then

$$z^d P_m(z^{-1}) = (-1)^m P_m(z), \quad \varepsilon_{d-n} = (-1)^m \varepsilon_n. \quad (7.4)$$

The cyclotomic factorization is

$$P_m(z) = (1 - z)^m \prod_{r=1}^{m-1} \Phi_{2^r}(z)^{m-r}, \quad (7.5)$$

where Φ_d denotes the d -th cyclotomic polynomial.

7.2 The infinite product and Mahler equation

For $|z| < 1$, coefficient stabilization gives

$$E(z) = \sum_{n \geq 0} \varepsilon_n z^n = \prod_{j \geq 0} (1 - z^{2^j}). \quad (7.6)$$

It satisfies the first-order Mahler equation

$$E(z) = (1 - z)E(z^2), \quad E(0) = 1. \quad (7.7)$$

The zero–one generating series is

$$T(z) = \sum_{n \geq 0} \tau(n) z^n = \frac{1}{2} \left(\frac{1}{1 - z} - E(z) \right). \quad (7.8)$$

7.3 A direct natural-boundary theorem

Theorem 7.2 (Natural boundary). *Added in this synthesis.* Every point of $|z| = 1$ is a singular point of analytic continuation of E . In particular, the unit circle is a natural boundary, and E is transcendental over $\mathbb{C}(z)$.

Proof. Let ζ be a dyadic root of unity, say $\zeta^{2^r} = 1$. For $0 < \rho < 1$, the factors with $j \geq r$ are real numbers in $[0, 1]$, and

$$|E(\rho\zeta)| \leq 2^r (1 - \rho^{2^r}).$$

Hence $E(\rho\zeta) \rightarrow 0$ radially as $\rho \uparrow 1$. Dyadic roots of unity are dense on the circle. If E continued through a nonempty arc, the continuation would vanish at infinitely many dyadic roots with an accumulation point in its domain. The identity theorem would force it to vanish identically, contradicting $E(0) = 1$. Thus no arc permits continuation. An algebraic function over $\mathbb{C}(z)$ has only finitely many branch points and poles on the Riemann sphere, so it cannot have a natural boundary. $\square$

Remark 7.3. The general rational-or-natural-boundary dichotomy for Mahler functions is due to Randé and has modern proofs by Bell, Coons, and Rowland [5]. Here the dense radial-zero argument is specific and self-contained.

8 The ruler sequence, coefficient recurrences, partitions, and determinants

Define the odd ruler values

$$a_r = 2^{\nu_2(r)+1} - 1 \quad (r \geq 1), \quad (8.1)$$

so $a_1, a_2, \dots = 1, 3, 1, 7, 1, 3, 1, 15, \dots$. Logarithmic expansion of (7.6) gives

$$\boxed{\log E(z) = - \sum_{r \geq 1} \frac{a_r}{r} z^r, \quad - \frac{zE'(z)}{E(z)} = \sum_{r \geq 1} a_r z^r.} \quad (8.2)$$

Indeed, the coefficient at z^r receives $-2^j/r$ from every $2^j \mid r$, whose sum is $-a_r/r$.

Comparing coefficients in $-zE' = E \sum a_r z^r$ yields a nonbinary convolution recurrence:

$$\boxed{n\varepsilon_n = - \sum_{r=1}^n a_r \varepsilon_{n-r} \quad (n \geq 1).} \quad (8.3)$$

The ruler sequence itself obeys

$$a_{2n+1} = 1, \quad a_{2n} = 2a_n + 1. \quad (8.4)$$

Its Lambert and Dirichlet series are

$$\sum_{r \geq 1} a_r z^r = \sum_{j \geq 0} \frac{2^j z^{2^j}}{1 - z^{2^j}}, \quad (8.5)$$

$$\sum_{r \geq 1} \frac{a_r}{r^s} = \boxed{\frac{\zeta(s)}{1 - 2^{1-s}}} \quad (\Re s > 1). \quad (8.6)$$

For the second formula, split $r = 2^k u$ with u odd and sum a geometric series.

8.1 Partition and Bell-polynomial formulas

Exponentiating (8.2) gives the finite partition sum

$$\boxed{\varepsilon_n = \sum_{\substack{k_1, \dots, k_n \geq 0 \\ k_1 + 2k_2 + \dots + nk_n = n}} \prod_{r=1}^n \frac{1}{k_r!} \left(-\frac{a_r}{r}\right)^{k_r}.} \quad (8.7)$$

The rational summands cancel to exactly ± 1 . In terms of complete exponential Bell polynomials,

$$\boxed{\varepsilon_n = \frac{1}{n!} \mathcal{B}_n(-0!a_1, -1!a_2, \dots, -(n-1)!a_n).} \quad (8.8)$$

8.2 A factorial determinant

For $n \geq 1$, let

$$H_n = \begin{pmatrix} a_1 & 1 & 0 & \cdots & 0 \\ a_2 & a_1 & 2 & \ddots & \vdots \\ a_3 & a_2 & a_1 & \ddots & 0 \\ \vdots & \vdots & \ddots & \ddots & n-1 \\ a_n & a_{n-1} & \cdots & a_2 & a_1 \end{pmatrix}. \quad (8.9)$$

Then

$$\varepsilon_n = \frac{(-1)^n}{n!} \det H_n, \quad |\det H_n| = n!. \quad (8.10)$$

To verify it, expand a lower-Hessenberg determinant to obtain

$$D_n = \sum_{r=1}^n (-1)^{r+1} \frac{(n-1)!}{(n-r)!} a_r D_{n-r}, \quad D_0 = 1,$$

and set $c_n = (-1)^n D_n / n!$; then c_n satisfies (8.3) with $c_0 = 1$.

9 Prouhet cancellation and a master finite-difference identity

9.1 Exact vanishing order

Every factor in (7.2) has a simple zero at $z = 1$, so

$$P_m(z) = (-1)^m 2^{\binom{m}{2}} (z-1)^m + \mathcal{O}((z-1)^{m+1}). \quad (9.1)$$

Thus P_m has a zero of exact order m at one.

Theorem 9.1 (Prouhet cancellation). *For every polynomial p of degree less than m ,*

$$\sum_{n=0}^{2^m-1} \varepsilon_n p(n) = 0. \quad (9.2)$$

Equivalently, the evil and odious subsets of $\{0, \dots, 2^m - 1\}$ have equal sums of like powers through degree $m - 1$.

Proof. The derivative $P_m^{(r)}(1)$ equals $\sum_{n < 2^m} \varepsilon_n n^r$ and vanishes for $r < m$. Falling factorials form a basis of the polynomial ring. $\square$

The first surviving moments are

$$\sum_{n < 2^m} \varepsilon_n n^m = (-1)^m m! 2^{\binom{m}{2}}, \quad (9.3)$$

$$\sum_{n < 2^m} \varepsilon_n \binom{n}{m} = (-1)^m 2^{\binom{m}{2}}. \quad (9.4)$$

9.2 The operator formula

For $h \in \mathbb{C}$, let $T_h f(x) = f(x + h)$ and $\Delta_h = T_h - I$. Expanding a product of commuting shifts proves the master identity

$$\sum_{n=0}^{2^m-1} \varepsilon_n f(x + nh) = \left[\prod_{j=0}^{m-1} (I - T_{2^j h}) f \right] (x) = (-1)^m \Delta_h \Delta_{2h} \cdots \Delta_{2^{m-1}h} f(x). \quad (9.5)$$

This single equation generates Prouhet cancellation, exact remainders, reciprocal-power positivity, and sparse Pascal analogues.

If $h \geq 0$ and $f \in C^m$, repeated use of the fundamental theorem of calculus gives

$$\sum_{n < 2^m} \varepsilon_n f(x + nh) = (-1)^m \int_0^h \int_0^{2h} \cdots \int_0^{2^{m-1}h} f^{(m)}(x + t_0 + \cdots + t_{m-1}) dt_{m-1} \cdots dt_0, \quad (9.6)$$

$$\left| \sum_{n < 2^m} \varepsilon_n f(x + nh) \right| \leq 2^{\binom{m}{2}} h^m \sup |f^{(m)}|. \quad (9.7)$$

Corollary 9.2 (Positive reciprocal-power sums). *For $a, s > 0$,*

$$\sum_{n=0}^{2^m-1} \frac{\varepsilon_n}{(a + n)^s} = (s)_m \int_0^1 \int_0^2 \cdots \int_0^{2^{m-1}} \frac{dt_{m-1} \cdots dt_0}{(a + t_0 + \cdots + t_{m-1})^{s+m}} > 0, \quad (9.8)$$

where $(s)_m = s(s+1) \cdots (s+m-1)$.

Proof. Use (9.6) with $f(x) = (a + x)^{-s}$. Its m -th derivative is $(-1)^m (s)_m (a + x)^{-s-m}$, so the signs cancel. $\square$

9.3 All higher moments

Set

$$A_m = \frac{2^m - 1}{2}, \quad c_{m,r} = \frac{B_{2r}}{2r(2r)!} \frac{2^{2rm} - 1}{2^{2r} - 1} \quad (r \geq 1), \quad (9.9)$$

where B_{2r} are Bernoulli numbers with $B_2 = 1/6$. Substituting $z = e^t$ in the finite product and using the expansion of $\log(\sinh x/x)$ gives

$$\sum_{n < 2^m} \varepsilon_n e^{nt} = (-1)^m 2^{\binom{m}{2}} t^m \exp \left(A_m t + \sum_{r \geq 1} c_{m,r} t^{2r} \right). \quad (9.10)$$

Consequently, for every $\ell \geq 0$,

$$\sum_{n < 2^m} \varepsilon_n n^{m+\ell} = (-1)^m 2^{\binom{m}{2}} (m + \ell)! [t^\ell] \exp \left(A_m t + \sum_{r \geq 1} c_{m,r} t^{2r} \right). \quad (9.11)$$

After centering at A_m , only even corrections survive:

$$\sum_{n < 2^m} \varepsilon_n e^{(n - A_m)t} = (-1)^m 2^{\binom{m}{2}} t^m \exp \left(\sum_{r \geq 1} c_{m,r} t^{2r} \right). \quad (9.12)$$

Hence

$$\sum_{n < 2^m} \varepsilon_n (n - A_m)^q = 0 \quad \text{unless } q \geq m \text{ and } q \equiv m \pmod{2}. \quad (9.13)$$

This also follows from the reciprocity (7.4).

For binomial moments, the exact coefficient generator is

$$\sum_{r \geq 0} \left(\sum_{n < 2^m} \varepsilon_n \binom{n}{r} \right) u^r = \prod_{j=0}^{m-1} \left(1 - (1+u)^{2^j} \right). \quad (9.14)$$

It displays the zeros for $r < m$ and the first value (9.4) without conversion between power bases.

10 Sparse Prouhet identities on arbitrary Pascal supports

Let

$$J(n) = \{j \geq 0 : b_j(n) = 1\}, \quad s = |J(n)| = \text{wt}(n), \quad \beta(n) = \sum_{j \in J(n)} j. \quad (10.1)$$

Theorem 10.1 (Pascal-support product). *For every $n \geq 0$,*

$$R_n(z) := \sum_{k \preceq 2n} \varepsilon_k z^k = \prod_{j \in J(n)} (1 - z^{2^j}). \quad (10.2)$$

Equivalently,

$$R_n(z) = \sum_{k=0}^n \varepsilon_k \mathbf{1}_{\binom{n}{k} \text{ odd}} z^k. \quad (10.3)$$

Proof. Expand the product. Each chosen subset of $J(n)$ gives one submask k and coefficient $(-1)^{\text{wt}(k)}$. Lucas's criterion gives the second form. $\square$

The polynomial R_n has a zero of order s at one. Therefore

$$\sum_{k \preceq 2n} \varepsilon_k p(k) = 0 \quad (\deg p < s), \quad (10.4)$$

and the first surviving moment is

$$\sum_{k \preceq 2n} \varepsilon_k k^s = (-1)^s s! 2^{\beta(n)}. \quad (10.5)$$

In particular, every nonzero Pascal row has equally many evil and odious indices among the positions of its odd entries.

The operator form is

$$\sum_{k \preceq 2n} \varepsilon_k f(x + hk) = \prod_{j \in J(n)} (I - T_{2^j h}) f(x). \quad (10.6)$$

For $f \in C^s$ and $h \geq 0$, its absolute value is at most

$$2^{\beta(n)} h^s \sup_{x \leq y \leq x+nh} |f^{(s)}(y)|. \quad (10.7)$$

Example 10.2. For $n = 13 = 1101_2$, the odd positions of Pascal row 13 are

$$0, 1, 4, 5, 8, 9, 12, 13.$$

Their signed sums against $1, k, k^2$ vanish, while

$$\sum_{k \preceq_2 13} \varepsilon_k k^3 = -3! 2^{0+2+3} = -192.$$

11 The bit series in characteristic two

In $\mathbb{F}_2[[x]]$, put

$$\Theta(x) = \sum_{n \geq 0} \tau(n) x^n. \quad (11.1)$$

Splitting even and odd coefficients gives

$$\Theta(x) = (1+x)\Theta(x^2) + \frac{x}{1+x^2}. \quad (11.2)$$

Frobenius says $\Theta(x^2) = \Theta(x)^2$, and $1+x^2 = (1+x)^2$. Hence

$$\boxed{(1+x)^3 \Theta(x)^2 + (1+x)^2 \Theta(x) + x = 0.} \quad (11.3)$$

With $Y(x) = (1+x)\Theta(x)$, this is the Artin–Schreier equation

$$\boxed{Y(x)^2 + Y(x) = \frac{x}{1+x}.} \quad (11.4)$$

Thus the same sequence whose complex generating function is transcendental is algebraic of degree two over $\mathbb{F}_2(x)$, an instructive contrast between characteristic zero and characteristic two.

12 Iterated prefix sums and terminal zero runs

Define inclusive discrete antiderivatives by

$$S_n^{(0)} = \varepsilon_n, \quad S_n^{(k+1)} = \sum_{j=0}^n S_j^{(k)}. \quad (12.1)$$

The first one is explicit:

$$S_{2q}^{(1)} = \varepsilon_q, \quad S_{2q+1}^{(1)} = 0. \quad (12.2)$$

Repeated summation gives

$$\boxed{S_n^{(k)} = \sum_{j=0}^n \binom{n-j+k-1}{k-1} \varepsilon_j,} \quad (12.3)$$

$$\boxed{\sum_{n \geq 0} S_n^{(k)} z^n = \frac{E(z)}{(1-z)^k}.} \quad (12.4)$$

Theorem 12.1 (Dyadic terminal zero run). *If $1 \leq k \leq r$, then*

$$S_{2^r-k+i}^{(k)} = 0 \quad (0 \leq i < k), \quad (12.5)$$

while

$$S_{2^r-k-1}^{(k)} = (-1)^{r-k}, \quad S_{2^r}^{(k)} = -1. \quad (12.6)$$

Proof. Below degree 2^r , replace $E(z)$ in (12.4) by $P_r(z)$. Since

$$\frac{P_r(z)}{(1-z)^k} = (1-z)^{r-k} \prod_{j=0}^{r-1} (1+z+\dots+z^{2^j-1})$$

is a polynomial of degree $2^r - k - 1$, the next k coefficients vanish. Its leading coefficient gives the first boundary value; the first omitted factor $1 - z^{2^r}$ gives the value at 2^r . $\square$

13 Walsh, ordinary Fourier, and cyclotomic formulas

13.1 The Boolean-cube character

For $0 \leq n < 2^m$, let $\mathbf{b}(n) \in \mathbb{F}_2^m$ be its digit vector and let $\mathbf{1} = (1, \dots, 1)$. Then

$$\varepsilon_n = (-1)^{\mathbf{1} \cdot \mathbf{b}(n)}. \quad (13.1)$$

Equivalently, in natural binary order,

$$(\varepsilon_0, \dots, \varepsilon_{2^m-1})^\top = \begin{pmatrix} 1 \\ -1 \end{pmatrix}^{\otimes m}. \quad (13.2)$$

Thus the finite block is one Walsh–Hadamard character. Orthogonality gives

$$\sum_{n < 2^m} \varepsilon_n (-1)^{\mathbf{a} \cdot \mathbf{b}(n)} = \begin{cases} 2^m, & \mathbf{a} = \mathbf{1}, \\ 0, & \mathbf{a} \neq \mathbf{1}. \end{cases} \quad (13.3)$$

The Walsh spectrum is a point mass even though the ordinary Fourier spectrum is spread over all odd frequencies.

For $x \in [0, 1)$, set $r_j(x) = (-1)^{\lfloor 2^{j+1}x \rfloor}$. If $n/2^m \leq x < (n+1)/2^m$, then

$$\varepsilon_n = \prod_{j=0}^{m-1} r_j(x). \quad (13.4)$$

13.2 Trigonometric factorization

Putting $z = e^{2\pi i x}$ in (7.2) gives

$$P_m(e^{2\pi i x}) = (-2i)^m e^{\pi i (2^m-1)x} \prod_{j=0}^{m-1} \sin(\pi 2^j x). \quad (13.5)$$

In particular,

$$|P_m(e^{2\pi i x})| = 2^m \prod_{j=0}^{m-1} |\sin(\pi 2^j x)|. \quad (13.6)$$

This is the finite Fourier counterpart of the sinc products that occur in the Fourier transform of the Rvachev up-function.

13.3 Exact discrete Fourier transform

For integer k , define

$$\widehat{\varepsilon}_m(k) = \sum_{n=0}^{2^m-1} \varepsilon_n e^{-2\pi i k n / 2^m}. \quad (13.7)$$

Then

$$\widehat{\varepsilon}_m(k) = \prod_{j=0}^{m-1} \left(1 - e^{-2\pi i k 2^j / 2^m} \right). \quad (13.8)$$

Every even mode vanishes. If k is odd, no factor vanishes and

$$\widehat{\varepsilon}_m(k) = (2i)^m e^{-\pi i k (1-2^{-m})} \prod_{r=1}^m \sin\left(\frac{\pi k}{2^r}\right). \quad (13.9)$$

Finite Fourier inversion therefore gives a cyclotomic formula for each term:

$$\varepsilon_n = 2^{-m} \sum_{\substack{0 \leq k < 2^m \\ k \text{ odd}}} \widehat{\varepsilon}_m(k) e^{2\pi i k n / 2^m} \quad (0 \leq n < 2^m). \quad (13.10)$$

The formula is independent of how large m is, once $2^m > n$.

14 Riesz products, autocorrelation, and spectral type

Define

$$g_m(x) = 2^{-m} |P_m(e^{2\pi i x})|^2. \quad (14.1)$$

Then

$$g_m(x) = \prod_{j=0}^{m-1} (1 - \cos(2\pi 2^j x)). \quad (14.2)$$

Parseval's identity gives $\int_0^1 g_m(x) dx = 1$, so $g_m(x) dx$ is a probability measure. Its weak limit is the classical Thue–Morse Riesz-product measure μ_{TM} .

For dyadic averages, put

$$\eta_m(r) = 2^{-m} \sum_{n=0}^{2^m-1} \varepsilon_n \varepsilon_{n+r}. \quad (14.3)$$

Even–odd splitting gives the exact recurrences

$$\eta_{m+1}(2r) = \eta_m(r), \quad (14.4)$$

$$\eta_{m+1}(2r+1) = -\frac{\eta_m(r) + \eta_m(r+1)}{2}. \quad (14.5)$$

For each fixed r , the limit $\eta(r) = \lim_m \eta_m(r)$ exists and is characterized by

$$\eta(0) = 1, \quad \eta(2r) = \eta(r), \quad \eta(2r+1) = -\frac{\eta(r) + \eta(r+1)}{2}. \quad (14.6)$$

For example,

$$\eta(1) = \eta(2) = -\frac{1}{3}, \quad \eta(3) = \frac{1}{3}, \quad \eta(5) = 0. \quad (14.7)$$

These are the Fourier coefficients of μ_{TM} .

Literature theorem 14.1 (Purely singular continuous spectrum). The Riesz-product measure μ_{TM} has no atoms and is singular with respect to Lebesgue measure. Accordingly, the balanced Thue–Morse substitution has a purely singular continuous spectral component associated with the weight $(1, -1)$; see Queffélec [17] and Baake–Grimm [4]. The recursion (14.6), the Riesz product (14.2), and the substitution dynamical system are three exact descriptions of the same measure.

Literature theorem 14.2 (Gelfond’s exponential-sum bound). There is an absolute constant C such that

$$\sup_{x \in \mathbb{R}} |P_m(e^{2\pi i x})| \leq C 3^{m/2}. \quad (14.8)$$

Equivalently, for arbitrary N ,

$$\sup_x \left| \sum_{n < N} \varepsilon_n e^{2\pi i n x} \right| \ll N^{\log 3 / \log 4}. \quad (14.9)$$

The exponent $\log 3 / \log 4$ is sharp. This estimate, originating in Gelfond’s work [9], is the analytic input behind many distribution results for digit sums.

15 Base- b Thue–Morse constants and Mahler transcendence

For an integer $b \geq 2$, define

$$\Theta_b = \sum_{n \geq 0} \frac{\tau(n)}{b^{n+1}}. \quad (15.1)$$

Substituting $z = 1/b$ in (7.8) gives

$$\Theta_b = \frac{1}{2(b-1)} - \frac{1}{2b} \prod_{j \geq 0} (1 - b^{-2^j}). \quad (15.2)$$

The signed companion is

$$\sum_{n \geq 0} \frac{\varepsilon_n}{b^{n+1}} = \frac{1}{b} \prod_{j \geq 0} (1 - b^{-2^j}). \quad (15.3)$$

For $b = 2$, $\Theta_2 = 0.011010011001 \dots_2$ is the classical Thue–Morse–Mahler constant.

Nonperiodicity already proves that Θ_b is irrational: a rational number has an eventually periodic base- b expansion, and the digits here are not eventually periodic. A much deeper statement is known.

Literature theorem 15.1 (Mahler). For every integer $b \geq 2$, the number Θ_b is transcendental. Mahler established this through the functional equation $E(z) = (1 - z)E(z^2)$; later proofs use automatic sequences, Ridout’s theorem, or modern Mahler-method machinery. See Mahler [12], Allouche–Shallit [2], and Bugeaud–Queffélec [6].

16 Shifted Dirichlet series and Mellin transforms

For $a > 0$, initially in $\Re s > 0$, define

$$D_a(s) = \sum_{n \geq 0} \frac{\varepsilon_n}{(n + a)^s}. \quad (16.1)$$

The convergence follows from the bounded prefix sums in section 6 and Dirichlet summation.

Theorem 16.1 (Entire continuation). *Added in this synthesis.* For every $a > 0$, $D_a(s)$ extends to an entire function of s . It satisfies

$$D_a(s) = 2^{-s} (D_{a/2}(s) - D_{(a+1)/2}(s)). \quad (16.2)$$

Proof. The functional equation follows by separating even and odd indices. For the continuation, group the series into blocks of length 2^m :

$$D_a(s) = 2^{-ms} \sum_{q \geq 0} \varepsilon_q \sum_{r=0}^{2^m-1} \varepsilon_r \left(q + \frac{a+r}{2^m} \right)^{-s}. \quad (16.3)$$

The inner sum is an m -fold mixed difference of x^{-s} ; uniformly for s in a compact set it is $\mathcal{O}(q^{-\Re s - m})$. Given any compact subset of the s -plane, choose m large enough to make the outer series normally convergent there. The representations agree where the original series converges, so they patch to an entire function. $\square$

There is also a Mellin representation:

$$D_a(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-at} E(e^{-t}) dt. \quad (16.4)$$

Initially this follows by termwise integration. Near zero, for every $m \geq 1$,

$$0 < E(e^{-t}) \leq 2^{\binom{m}{2}} t^m, \quad (16.5)$$

using the first m factors of the product. This super-polynomial flatness also explains the entire continuation in (16.4).

At $s = 1$, write

$$H(a) = D_a(1) = \sum_{n \geq 0} \frac{\varepsilon_n}{n+a}. \quad (16.6)$$

Then

$$H(a) = \int_0^1 x^{a-1} E(x) dx > 0, \quad (16.7)$$

and

$$H(a) = \frac{1}{2} H(a/2) - \frac{1}{2} H((a+1)/2). \quad (16.8)$$

The positivity is nonobvious in the alternating series but immediate in the integral because every factor of $E(x)$ is positive for $0 < x < 1$.

17 A two-parameter product calculus

For $a, b > 0$, define

$$G(a, b) = \prod_{n \geq 0} \left(\frac{n+a}{n+b} \right)^{\varepsilon_n}. \quad (17.1)$$

The logarithm converges by summation by parts: the partial sums of ε_n are bounded and the first difference of $\log((n+a)/(n+b))$ is summable. Directly,

$$G(a, a) = 1, \quad G(a, b)G(b, c) = G(a, c), \quad G(b, a) = G(a, b)^{-1}. \quad (17.2)$$

Pairing even and odd indices yields

$$G(a, b) = \frac{G(a/2, b/2)}{G((a+1)/2, (b+1)/2)}. \quad (17.3)$$

17.1 The Woods–Robbins product

The most famous evaluation is

$$G\left(\frac{1}{2}, 1\right) = \prod_{n \geq 0} \left(\frac{2n+1}{2n+2}\right)^{\varepsilon_n} = \frac{1}{\sqrt{2}}. \quad (17.4)$$

For a short proof, put

$$A = \prod_{n \geq 0} \left(\frac{2n+1}{2n+2}\right)^{\varepsilon_n}, \quad B = \prod_{n \geq 1} \left(\frac{2n}{2n+1}\right)^{\varepsilon_n}.$$

Splitting the product AB into even and odd indices gives $AB = \frac{1}{2}A^{-1}B$, hence $A^2 = 1/2$, and positivity selects the positive root.

A second closed value, proved in the modern product calculus, is

$$G\left(\frac{1}{4}, \frac{3}{4}\right) = \frac{1}{2}. \quad (17.5)$$

See Riasat [18] and Allouche–Riasat–Shallit [3] for systematic extensions.

17.2 Three dyadic block-product limits

The product evaluations imply

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} \left(k + \frac{1}{2}\right)^{\varepsilon_k} = \frac{1}{2}, \quad (17.6)$$

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} (k+1)^{\varepsilon_k} = \frac{1}{\sqrt{2}}, \quad (17.7)$$

$$\lim_{m \rightarrow \infty} \prod_{k=0}^{2^m-1} (k+1)^{(-1)^k \varepsilon_k} = \frac{1}{2\sqrt{2}}. \quad (17.8)$$

For example, pairing adjacent factors in the first two finite products gives

$$\begin{aligned} \prod_{k < 2^m} \left(k + \frac{1}{2}\right)^{\varepsilon_k} &= \prod_{j < 2^{m-1}} \left(\frac{j+1/4}{j+3/4}\right)^{\varepsilon_j}, \\ \prod_{k < 2^m} (k+1)^{\varepsilon_k} &= \prod_{j < 2^{m-1}} \left(\frac{j+1/2}{j+1}\right)^{\varepsilon_j}. \end{aligned}$$

The third limit is the product of the first two. Dyadic truncation is natural because it preserves the exact self-similar blocks.

18 Combinatorics on the infinite word

18.1 Uniform recurrence and reversal symmetry

The substitution matrix of $\mu(0) = 01, \mu(1) = 10$ is strictly positive: each image contains both letters. More concretely, every factor occurring in the fixed point occurs in some finite word $\mu^k(0)$; every

sufficiently large supertile contains copies of both $\mu^k(0)$ and $\mu^k(1)$. Therefore every factor recurs with bounded gaps.

Proposition 18.1 (Uniform recurrence). *The Thue–Morse word is uniformly recurrent: for every finite factor w , there exists $L(w)$ such that every factor of length $L(w)$ contains an occurrence of w .*

The reciprocity (7.4) gives a precise reversal law for the finite blocks. If $0 \leq n < 2^m$, then

$$\varepsilon_{2^m-1-n} = (-1)^m \varepsilon_n. \quad (18.1)$$

Thus a level- m signed block is a palindrome when m is even and the negative of its reversal when m is odd. In zero–one notation, odd levels are complement-palindromes.

18.2 An elementary linear complexity bound

Let $p(\ell)$ denote the number of distinct factors of length ℓ . Choose k minimal with $2^k \geq \ell$. Every length- ℓ factor lies inside two consecutive level- k supertiles $\mu^k(a)\mu^k(b)$, where $a, b \in \{0, 1\}$. There are at most four choices for ab and at most 2^k starting offsets. Hence

$$p(\ell) \leq 4 \cdot 2^k < 8\ell \quad (\ell \geq 2). \quad (18.2)$$

Sharper exact piecewise-linear formulas are known, but this elementary bound already has two consequences: the topological entropy is zero, and the word is not normal. Indeed, a normal binary word contains all 2^ℓ words of length ℓ , incompatible with linear complexity for large ℓ .

Literature theorem 18.2 (Thue’s overlap theorem). The infinite Thue–Morse word is overlap-free: it contains no factor of the form $axaxa$, where a is a letter and x a possibly empty word. In particular it contains no cube www . This was one of Thue’s foundational results in combinatorics on words [23]; modern treatments appear in Allouche–Shallit [2]. The statement is about factors of the original word and should not be confused with the subsequence $\tau(n^3)$ discussed later.

19 Distribution in arithmetic and polynomial subsequences

The deterministic word has low factor complexity, yet arithmetically sampled subsequences can display strong pseudorandomness. The following theorems are not reproved here; they locate the exact identities of the atlas inside modern analytic number theory.

Literature theorem 19.1 (Newman’s phenomenon). Let

$$f_{3,0}(N) = \sum_{0 \leq n < N} \varepsilon_{3n}. \quad (19.1)$$

Then $f_{3,0}(N) > 0$ for every $N \geq 1$: among the first N multiples of three, evil numbers always outnumber odious numbers. Newman proved the quantitative bounds

$$\frac{1}{20} N^{\log 3 / \log 4} \leq f_{3,0}(N) \leq 5 N^{\log 3 / \log 4}, \quad (19.2)$$

and Coquet later described the exact fractal fluctuation [15, 7, 20].

Literature theorem 19.2 (Prime indices). Mauduit and Rivat proved that binary digit-sum parity is equidistributed along the primes. In signed form,

$$\sum_{p \leq x} \varepsilon_p = o(\pi(x)), \quad (19.3)$$

so asymptotically half the primes are evil and half odious [13].

Literature theorem 19.3 (Squares are normal). Drmota, Mauduit, and Rivat proved that the subsequence $(\tau(n^2))_{n \geq 0}$ is normal: every binary word of length L occurs with limiting frequency 2^{-L} [8]. This is far stronger than the one-bit balance $\sum_{n < N} \varepsilon_{n^2} = o(N)$.

Literature theorem 19.4 (Cubes are simply normal). Spiegelhofer proved

$$\#\{n < N : \tau(n^3) = 0\} \sim \frac{N}{2}, \quad \text{equivalently} \quad \sum_{n < N} \varepsilon_{n^3} = o(N). \quad (19.4)$$

As of the cited result, full block normality along cubes was not asserted; the theorem establishes one-symbol balance and resolves the cubic parity case of a longstanding Gelfond problem [21].

Literature theorem 19.5 (Level of distribution). Spiegelhofer proved a Bombieri–Vinogradov-type theorem with level of distribution one (up to the endpoint) for the Thue–Morse sequence. Among its applications, $(\tau(\lfloor n^c \rfloor))$ is simply normal for $1 < c < 2$ [22]. This illustrates how the finite Fourier product and carry control feed into arithmetic-progressions estimates.

20 Base- B root-of-unity generalization

Let $B \geq 2$, let $s_B(n)$ be the base- B digit sum, and let $\zeta \neq 1$ be a B -th root of unity. Define

$$u_n = \zeta^{s_B(n)}. \quad (20.1)$$

Then

$$u_{Bn+d} = \zeta^d u_n \quad (0 \leq d < B). \quad (20.2)$$

The finite generating polynomial is

$$P_{m,B,\zeta}(z) = \sum_{n < B^m} u_n z^n = \prod_{j=0}^{m-1} \left(\sum_{d=0}^{B-1} \zeta^d z^{dB^j} \right). \quad (20.3)$$

Because $\zeta^B = 1$, each factor can also be written as

$$\sum_{d=0}^{B-1} \zeta^d z^{dB^j} = \frac{1 - z^{B^{j+1}}}{1 - \zeta z^{B^j}}. \quad (20.4)$$

At $z = 1$, every factor has a simple zero and

$$\left. \frac{d}{dz} \sum_{d=0}^{B-1} \zeta^d z^{dB^j} \right|_{z=1} = -\frac{B^{j+1}}{1 - \zeta}. \quad (20.5)$$

Therefore $P_{m,B,\zeta}$ has a zero of exact order m , and the base- B Prouhet theorem follows.

Theorem 20.1 (Root-of-unity Prouhet hierarchy). *For every polynomial p with $\deg p < m$,*

$$\sum_{n=0}^{B^m-1} \zeta^{s_B(n)} p(n) = 0. \quad (20.6)$$

The first surviving moment is

$$\sum_{n=0}^{B^m-1} \zeta^{s_B(n)} n^m = (-1)^m m! \frac{B^{m(m+1)/2}}{(1 - \zeta)^m}. \quad (20.7)$$

Proof. Differentiate (20.3) at one as in the binary proof. The leading coefficient is the product of (20.5). Falling factorials again form a basis. $\square$

Fourier inversion over the cyclic group of B -th roots shows that the classes

$$\{0 \leq n < B^m : s_B(n) \equiv r \pmod{B}\}, \quad 0 \leq r < B, \quad (20.8)$$

have equal sums of like powers through degree $m - 1$. The binary Thue–Morse Prouhet partition is exactly $B = 2, \zeta = -1$.

For a prime p , Lucas’s theorem similarly gives a Pascal-row extension. If $n = \sum n_j p^j$, then

$$\# \left\{ 0 \leq k \leq n : \binom{n}{k} \not\equiv 0 \pmod{p} \right\} = \prod_j (n_j + 1). \quad (20.9)$$

Thus the binary count $2^{\text{wt}(n)}$ is the $p = 2$ specialization of a full digitwise product.

21 The discrete bridge to the Fabius and up-functions

The appearance of Thue–Morse in the Fabius-function development is governed by one algebraic pipeline:

binary character $\implies P_r(z) = \prod_{j < r} (1 - z^{2^j}) \implies \frac{P_r(z)}{(1 - z)^k} \implies \text{iterated-prefix coefficients.}$

(21.1)

The last arrow is exactly (12.4). The factor $(1 - z)^r \mid P_r(z)$ is the algebraic source of the terminal zero runs and of the high-order flatness that survives after scaling.

On the Fourier side, a normalized factor

$$\frac{1 - e^{-ih\xi}}{ih\xi} = e^{-ih\xi/2} \text{sinc}(h\xi/2)$$

turns a finite difference into a sinc factor. Iterating over dyadic steps produces finite products converging to

$$\Phi(\xi) = \prod_{j \geq 0} \text{sinc}\left(\frac{\pi\xi}{2^j}\right), \quad (21.2)$$

up to the conventional indexing and Fourier normalization. This is the Fourier transform of a compactly supported infinite convolution of uniform measures, namely the probability law underlying the Rvachev up-function. The sign identity (3.7) then explains, in a particularly literal way, why Thue–Morse signs are encoded in the same sinc product.

After normalization by $2^{\binom{k}{2}}$, coefficient grids built from $S_n^{(k)}$, together with suitable polygonal or spline interpolation, converge to the Fabius/up profile in the companion development. The exact normalization is already visible in the volume factor in (9.6) and in the first surviving moment (9.3). The Prouhet zero supplies flatness, while the Fourier product supplies compact support and smoothness. These are not two separate coincidences; they are the coefficient-space and frequency-space faces of the same product P_r .

Remark 21.1. The present atlas records the exact algebraic bridge rather than repeating the quantitative convergence analysis developed elsewhere in the Fabius-function project. The relevant repository modules named in the source drafts are:

ThueMorsePrefix.lean and ThueMorseGenerating.lean.

22 Representation comparison and a Lean roadmap

Representation	Primitive data	Especially effective for
Binary recursion	parity of appended digit	evaluation, induction, blocks
Two-kernel automaton	two states and digit matrices	automaticity, extraction
XOR character	Boolean group $\mathbb{F}_2^m$	Walsh analysis, Hamming spheres
Valuation	$\nu_2(n!)$, Kummer carries	binomial/Catalan arithmetic
Pascal support	Lucas submask criterion	log-free and sparse identities
Finite Euler product	independent digit choices	Prouhet, DFT, reciprocity
Ruler logarithm	divisors $2^j \mid n$	convolution, Bell, determinant
Finite difference	commuting translations	remainders, bounds, positivity
Riesz product	squared Fourier modulus	correlation and spectral measure
Mahler equation	scale map $z \mapsto z^2$	constants, transcendence, boundary
Prefix quotient	multiplication by $(1 - z)^{-k}$	discrete integration, Fabius

A dependency-respecting Lean formalization can proceed in the following order.

1. Prove the sign normalization and the two recurrences (2.1).
2. Establish block multiplicativity, the exact two-kernel, XOR multiplicativity, and prefix discrepancy.
3. Import or prove Legendre, Lucas, and Kummer specializations; derive the central-binomial and Catalan formulas.
4. Define P_m , prove the finite product, reciprocity, and the order of vanishing at one.
5. Formalize the shift-operator identity, Prouhet cancellation, the first surviving moment, and the sparse Pascal-support version.
6. Develop the logarithmic ruler recurrence; Bell polynomials and the Hessenberg determinant can be later corollaries.
7. Add finite DFT and Walsh transforms over $\mathbb{Z}/2^m\mathbb{Z}$ and $\mathbb{F}_2^m$.
8. Connect repeated prefix sums to $E(z)/(1 - z)^k$ and prove terminal zero runs before tackling analytic convergence to Fabius.

Likely theorem names should distinguish the bit τ , the sign ε , and the weight wt ; many informal sign errors arise from silently moving between these normalizations.

A The first 64 terms

The following table displays $\tau(n)$. Replacing 0 by +1 and 1 by −1 gives ε_n .

n	successive values							
0–7	0	1	1	0	1	0	0	1
8–15	1	0	0	1	0	1	1	0
16–23	1	0	0	1	0	1	1	0
24–31	0	1	1	0	1	0	0	1
32–39	1	0	0	1	0	1	1	0
40–47	0	1	1	0	1	0	0	1
48–55	0	1	1	0	1	0	0	1
56–63	1	0	0	1	0	1	1	0

Each row of length eight is a copy or complement of the first row according to the sign of its block index, as predicted by (2.4).

B A compact identity index

$$\begin{aligned}
 \tau(n) &\longleftrightarrow \varepsilon_n = 1 - 2\tau(n), \\
 \varepsilon_n &\longleftrightarrow (-1)^{\text{wt}(n)}, \\
 \text{wt}(n) &\longleftrightarrow n - \nu_2(n!), \\
 \text{wt}(n) &\longleftrightarrow \nu_2\binom{2n}{n}, \\
 b_j(n) &\longleftrightarrow \binom{n}{2^j} \bmod 2, \\
 k \preceq_2 n &\longleftrightarrow \binom{n}{k} \text{ odd}, \\
 \varepsilon_n &\longleftrightarrow [z^n] \prod_{j \geq 0} (1 - z^{2^j}), \\
 \sum_{n < 2^m} \varepsilon_n f(n) &\longleftrightarrow (-1)^m \Delta_1 \Delta_2 \cdots \Delta_{2^{m-1}} f(0), \\
 \eta(r) &\longleftrightarrow \widehat{\mu}_{\text{TM}}(r), \\
 S_n^{(k)} &\longleftrightarrow [z^n] \frac{E(z)}{(1-z)^k}.
 \end{aligned}$$

C Executable checks

The following short Python program checks a representative cross-section of the identities. It deliberately uses integer arithmetic except in the Fourier test.

```

from math import comb, factorial
import cmath

def wt(n: int) -> int:
    return n.bit_count()

def eps(n: int) -> int:
    return -1 if wt(n) & 1 else 1

def v2(n: int) -> int:
    if n <= 0:
        raise ValueError("v2 expects a positive integer")
    return (n & -n).bit_length() - 1

def catalan(n: int) -> int:

```

```

    return comb(2*n, n) // (n + 1)

for n in range(1, 500):
    assert v2(comb(2*n, n)) == wt(n)
    assert v2(catalan(n)) == wt(n + 1) - 1

for m in range(1, 9):
    # Prouhet moments.
    for r in range(m):
        assert sum(eps(n) * n**r for n in range(2**m)) == 0
    assert sum(eps(n) * n**m for n in range(2**m)) == \
        (-1)**m * factorial(m) * 2**(m*(m-1)//2)

    # DFT: every even frequency vanishes.
    for k in range(0, 2**m, 2):
        s = sum(eps(n) * cmath.exp(-2j*cmath.pi*k*n/2**m)
                for n in range(2**m))
        assert abs(s) < 1e-8

for n in range(1, 200):
    odd_positions = [k for k in range(n + 1) if comb(n, k) & 1]
    assert len(odd_positions) == 2**wt(n)
    assert sum(eps(k) for k in odd_positions) == 0

print("all checks passed")

```

D Bibliographic notes

The Thue–Morse sequence has been rediscovered in number theory, combinatorics on words, dynamics, harmonic analysis, automata, and fair allocation. The bibliography below is selective: it emphasizes sources that support the formula families merged here and the extra material added in this version. The four repository drafts themselves contain further specialized references.

References

- [1] J.-P. Allouche and J. Shallit, *The ubiquitous Prouhet–Thue–Morse sequence*, in *Sequences and Their Applications*, Springer, 1999, pp. 1–16.
- [2] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [3] J.-P. Allouche, S. Riasat, and J. Shallit, More infinite products involving the Thue–Morse sequence, *J. Integer Sequences* 22 (2019), Article 19.2.3; see also arXiv:1709.03398.
- [4] M. Baake and U. Grimm, The singular continuous diffraction measure of the Thue–Morse chain, *J. Phys. A* 41 (2008), 422001.
- [5] J. P. Bell, M. Coons, and E. Rowland, The rational–transcendental dichotomy of Mahler functions, *J. Integer Sequences* 16 (2013), Article 13.2.10.
- [6] Y. Bugeaud and M. Queffélec, On the rational approximation to the binary Thue–Morse–Mahler number, *J. Integer Sequences* 16 (2013), Article 13.2.1.

- [7] J. Coquet, A summation formula related to the binary digits, *Invent. Math.* 73 (1983), 107–115.
- [8] M. Drmota, C. Mauduit, and J. Rivat, The Thue–Morse sequence along the squares is normal, preprint dated 2013; published version in the analytic-number-theory literature.
- [9] A. O. Gelfond, Sur les nombres qui ont des propriétés additives et multiplicatives données, *Acta Arith.* 13 (1967/68), 259–265.
- [10] E. E. Kummer, Über die Ergänzungssätze zu den allgemeinen Reciprocitätsgesetzen, *J. Reine Angew. Math.* 44 (1852), 93–146.
- [11] É. Lucas, Théorie des fonctions numériques simplement périodiques, *Amer. J. Math.* 1 (1878), 184–240 and 289–321.
- [12] K. Mahler, Arithmetische Eigenschaften der Lösungen einer Klasse von Funktionalgleichungen, *Math. Ann.* 101 (1929), 342–366; corrigendum 103 (1930), 532.
- [13] C. Mauduit and J. Rivat, Sur un problème de Gelfond: la somme des chiffres des nombres premiers, *Ann. of Math.* 171 (2010), 1591–1646.
- [14] M. Morse, Recurrent geodesics on a surface of negative curvature, *Trans. Amer. Math. Soc.* 22 (1921), 84–100.
- [15] D. J. Newman, On the number of binary digits in a multiple of three, *Proc. Amer. Math. Soc.* 21 (1969), 719–721.
- [16] E. Prouhet, Mémoire sur quelques relations entre les puissances des nombres, *C. R. Acad. Sci. Paris* 33 (1851), 225.
- [17] M. Queffélec, *Substitution Dynamical Systems – Spectral Analysis*, 2nd ed., Lecture Notes in Mathematics 1294, Springer, 2010.
- [18] S. Riasat, Infinite products involving binary digit sums, arXiv:1709.04104, version of 2018.
- [19] D. Robbins, Solution to Problem E2692, *Amer. Math. Monthly* 86 (1979), 394–395.
- [20] J. Shallit, Rarefied Thue–Morse sums via automata theory and logic, arXiv:2302.09436, 2023.
- [21] L. Spiegelhofer, Thue–Morse along the sequence of cubes, arXiv:2308.09498, 2023.
- [22] L. Spiegelhofer, The level of distribution of the Thue–Morse sequence, *Compos. Math.* 156 (2020), 2560–2587.
- [23] A. Thue, Über unendliche Zeichenreihen, *Norske Vid. Selsk. Skr. I. Mat.-Nat. Kl.* 7 (1906), 1–22.
- [24] D. R. Woods, Problem E2692, *Amer. Math. Monthly* 85 (1978), 48.
- [25] V. Reshetnikov, PROVEIt: Analysis/FabiusFunction, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction>, accessed 26 August 2026.